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Transforming Offshore Operations Through IOCC Digitalization: A Centralized Digital Platform for Enhanced Collaboration, Visibility, and Operational Intelligence

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Abstract

The upstream oil and gas industry is experiencing a profound digital shift, driven by the need to enhance operational efficiency, ensure safety, and meet sustainability expectations. PTT Exploration and Production (PTTEP) has undertaken a strategic transformation through its Integrated Operations Control Center (IOCC), which centralizes offshore operations to be fully controlled and managed at PTTEP headquarters in Bangkok. More than a relocation of control system, crews and infrastructures, this initiative establishes a new operational paradigm that integrates people, processes, and technologies into a unified digital ecosystem.

The IOCC digitalization is designed on a cloud-native architecture with scalable data pipelines, analytics engines, and interactive digital workspaces. By aggregating data from production systems, field instrumentation, maintenance records, and AI-driven optimization models, the platform provides a consolidated environment for decision-making. Multi-level dashboards present production summaries, asset performance, and field-level operational insights, enabling stakeholders from executives to frontline engineers to align on shared objectives. Workspaces further extend functionality by supporting gas and condensate management, CO_2 membrane optimization, well configuration planning, and equipment condition monitoring. These features embed advisory logic, customizable visualization, and workflow automation to reduce manual effort, streamline collaboration, and accelerate the decision cycle.

Early results demonstrate measurable benefits, including enhanced oversight of production operations, improved wellhead control, and reduced hydrocarbon losses through optimized CO_2 membrane performance. Cross-asset collaboration has also been strengthened across seven production fields and three floating storage units, while operational reporting and planning processes have become more efficient and transparent. Importantly, the IOCC initiative is projected to deliver annual value realization of approximately USD 3.2 million, derived from productivity gains, reduced downtime, OPEX saving and better resource allocation.

Beyond technical advances, the IOCC fosters a cultural transformation by breaking down silos and instilling a proactive, data-driven mindset across multidisciplinary teams. This paper outlines the

development methodology, phased rollout, and initial outcomes, while also providing recommendations for sustaining long-term value through predictive analytics, digital twin integration, and advanced AI capabilities. The IOCC initiative represents a replicable model for energy companies seeking to accelerate digital transformation, elevate collaboration, and unlock sustainable operational intelligence in offshore environments.

Introduction

In recent years, the energy sector has undergone a significant paradigm shift, embracing digital transformation as a mean to drive greater efficiency, enhance safety, and meet growing expectations for environmental sustainability. Within the upstream oil and gas domain, this evolution is particularly pronounced, as operators seek to harness data-driven technologies to navigate increasingly complex operational landscapes. PTTEP's Integrated Operations Control Center (IOCC) initiative is a prime example of this transformation into action.

Rather than managing offshore assets through isolated support team and field-based control rooms where data flow is isolated and limited collaboration, the IOCC consolidates operational oversight into a unified, centrally located command center at PTTEP headquarters. This initiative is not merely a change in geography, it marks a foundational shift in how operations are conducted, monitored, and optimized. By eliminating fragmented workflows and siloed decision-making, the IOCC facilitates real-time situational awareness and more coordinated cross-functional responses across PTTEP's Gulf of Thailand (GoT) assets base.

At its core, the IOCC embodies a strategic commitment to intelligent operations. Through the seamless integration of digital workspaces, dynamic collaboration dashboards, and real-time analytics, the center enables a higher level of visibility, faster issue diagnosis, synchronized responses, dispersed expertise to converge in real time and agility across all operational domains. It is a forward-looking platform designed not just to improve daily operations, but to empower teams with the insights and tools necessary to anticipate challenges, respond proactively, and continuously elevate the performance and sustainability of offshore activities.

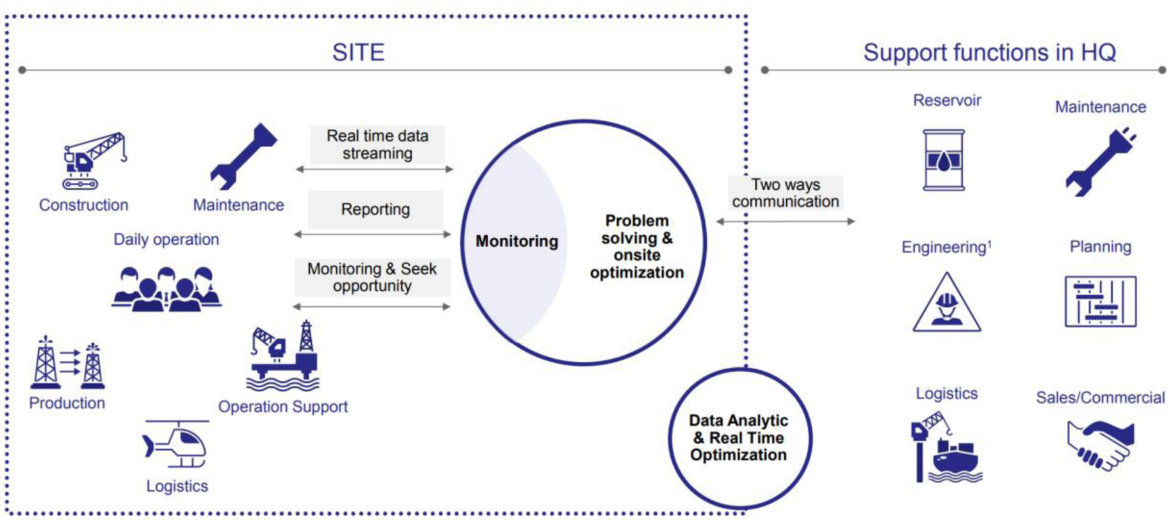

Figure 1—Traditional optimization relying on site and operation support

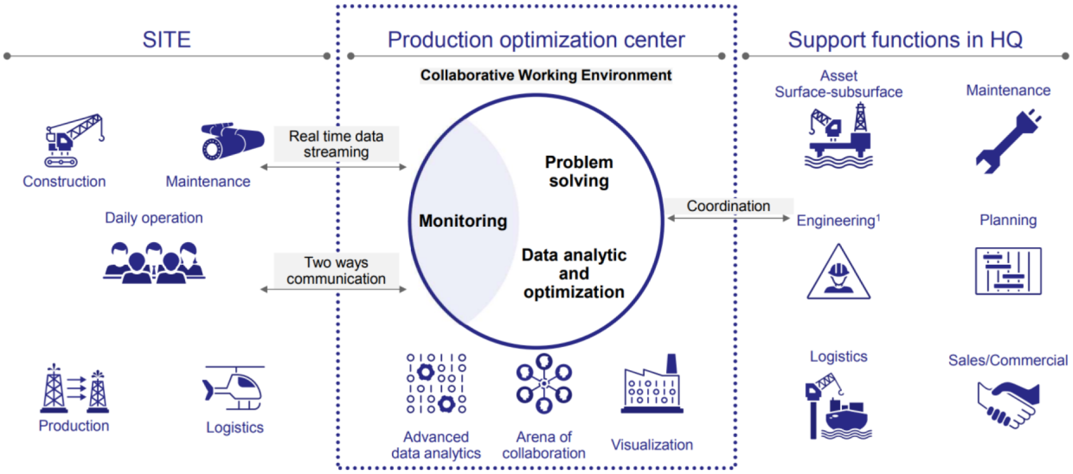

Figure 2—Optimization via centralized digital platform with shared resources enabled by PTTEP's IOCC digitalization

Methodology: Data Transformation for Operational Decision Making

The IOCC Digitalization project officially commenced in Q3 2024, guided by a structured yet agile development strategy. To accelerate progress and maintain close alignment with user needs, PTTEP established a dedicated in-house development squad that adopted Agile and SCRUM methodologies. This iterative approach allowed for continuous feedback loops, adaptive planning, and incremental delivery—ensuring the rapid evolution of key digital tools such as the IOCC Collaboration Dashboard and the Digital Workspace.

These platforms were designed from the ground up on a modern, cloud-native architecture as figure 3, emphasizing scalability, security, and high availability. By leveraging enterprise-grade infrastructure, the solution ensures robust performance under real-time operational demands. The system architecture incorporates a range of backend components—including real-time data ingestion pipelines, centralized data lakes, and powerful analytics engines—capable of processing both structured and semi-structured data from diverse sources across PTTEP's offshore operations. Together, these elements form the digital backbone of the IOCC, enabling seamless data integration and insight-driven decision-making at scale.

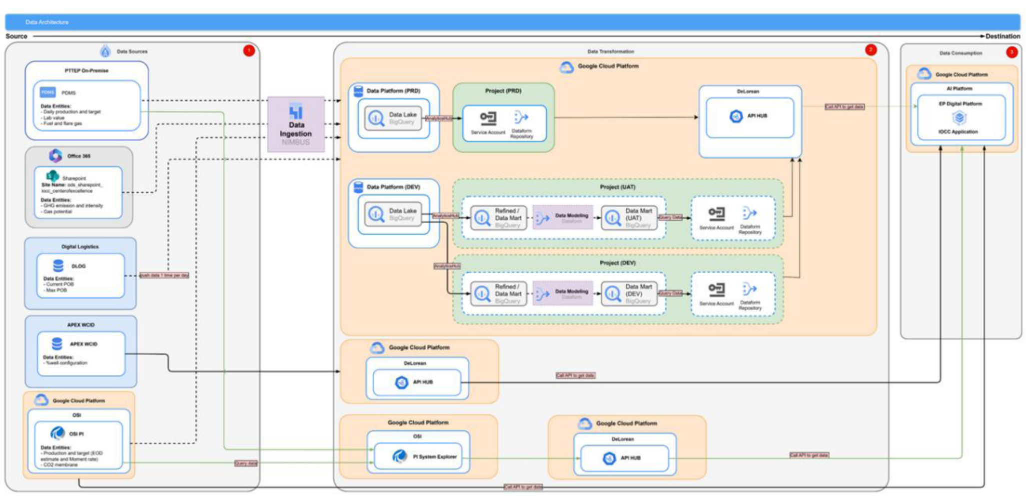

Figure 3—Data architecture of the IOCC digitalization, *(1) Data sources from field sensors, production systems, and enterprise applications; (2) Ingestion pipelines include data lake, data mart, and the centralized data platform; (3) The IOCC application layer, real-time analytics and AI models, visualization modules*.

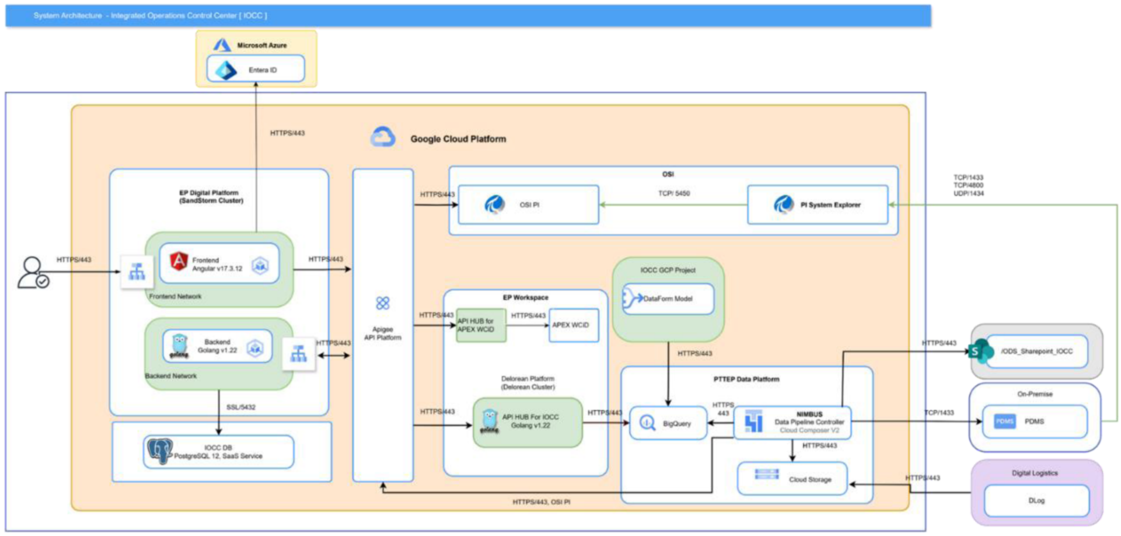

Figure 4—Digital Ecosystem of the IOCC Collaboration Dashboard and Digital Workspace.

Importantly, the IOCC tools are deeply integrated into PTTEP's broader digital ecosystem, providing seamless access to a wide range of operational data—including field instrumentation, log sheets, lab analyses, maintenance records, AI-based optimization models, and analytics tools. This integration forms the backbone of a data-centric operational culture, where information is not siloed within individual functions but is interconnected in a way that allows cross-domain insights to emerge naturally. It ensures that information generated in one operational context—such as a field sensor reading indicating a potential integrity issue—can be correlated with other relevant data sources, such as historical maintenance patterns or environmental conditions, in near real-time. As a result, decisions are not just made faster but are also made with greater confidence and alignment across teams. This holistic integration creates a unified knowledge base that encourages transparency, shortens feedback loops, and reinforces accountability across multiple functions including production, reservoir management, facilities engineering, and SHE. To turn raw data into actionable insights, the system applies three key mechanisms:

Visualization and Pre-Calculation

Operational data—whether structured (e.g., pressure and temperature readings) or unstructured (e.g., logbook entries or free-text lab notes)—often arrives in formats that are not immediately usable for decision-making. To make such data meaningful, the system performs real-time or near-real-time pre-processing through contextualized calculations, normalizations, unit harmonizations, and cross-timeframe aggregations. This is followed by transformation into visual representations that are intuitive yet powerful, such as heat maps, stacked line graphs, or trend-based alert icons. The goal is to enable the user to immediately grasp operational behavior, identify performance deviations, or detect temporal anomalies—without having to sort through raw data manually or cross-reference multiple spreadsheets. This mechanism increases the speed and clarity of initial assessment, serving both situational awareness and deeper pattern recognition.

Thresholds and Advisory Logic

To bridge the gap between data and understanding, the system embeds expert knowledge directly into its logic layer. Instead of showing absolute numbers alone, the platform highlights contextual meaning by applying thresholds, limits, and advisory flags that are pre-defined by engineering experts or derived from regulatory standards and best practices. These thresholds may include, for example, minimum required corrosion inhibitor dosage per wellhead platform, acceptable gas specifications per contract, or red flags for sudden spikes in GHG intensity. Once the system detects a breach or trend toward a limit, it doesn't just notify the user—it explains why this matters and often suggests corrective action or directs users to related data points for deeper analysis. This approach significantly reduces the mental burden on operators and analysts, allowing them to focus more on action and collaboration than interpretation.

Customizability and User Preferences

Given the wide array of users—ranging from field supervisors and production engineers to asset managers and executive-level decision-makers, the platform is designed with flexibility at its core. Every user or team can configure their dashboards to reflect the dimensions they care about most, such as asset grouping, timeframe, unit of measure, or KPI emphasis. Not only can charts and alerts be filtered and rearranged, but also entire dashboards can be saved as templates for recurring use or for quick reference during recurring meetings. This modular approach to UI/UX ensures relevance without compromising standardization. It also empowers users to "own" their digital space within the IOCC, increasing both adoption and accountability. The customization doesn't end at visualization—it extends into alert preferences, refresh intervals, data granularity levels, and even scenario planning modes for simulation or forecasting.

Platform Features and Functionality.

IOCC Homepage for essential features to support integrated operations and provide a holistic view of PTTEP's GoT Fields.

The homepage acts as a gateway or functional hub for operational tools, allowing users to easily navigate and access various functions and operational applications tailored to their needs. This includes **Collaboration Dashboards**: These dashboards likely provide aggregated insights and performance indicators for various operational areas, facilitating collaborative decision-making and performance monitoring. **Workspace**: This feature suggests a personalized or team-oriented workspace where users can organize their tasks, perform engineering simulations, work with digital optimization tools and access relevant documents. **Other Operational Applications**: The homepage provides also access to a suite of other applications crucial for daily operations, streamlining workflows and enhancing user experience by acting as a single point of entry. In addition, the homepage includes some real-time data from site such as 1) Production Rate: Users can view the current gas and condensate production rates in each field, offering an up-to-the-minute understanding of operational output. This immediate access to production metrics is vital for quick assessment and response. 2) Weather Conditions: Integration of real-time weather data, i.e. wave height and wind speed, helps in assessing environmental impacts on offshore operations, enabling proactive measures for safety and efficiency. And 3) CCTV Integration: Live CCTV feeds provide visual confirmation of the real conditions offshore, allowing control rooms panel operators who sit in IOCC to virtually monitor physical sea conditions. This functionality could be implemented with additional advanced features such as camera management, click-on-map functionality for camera positioning, and ensuring comprehensive visual monitoring.

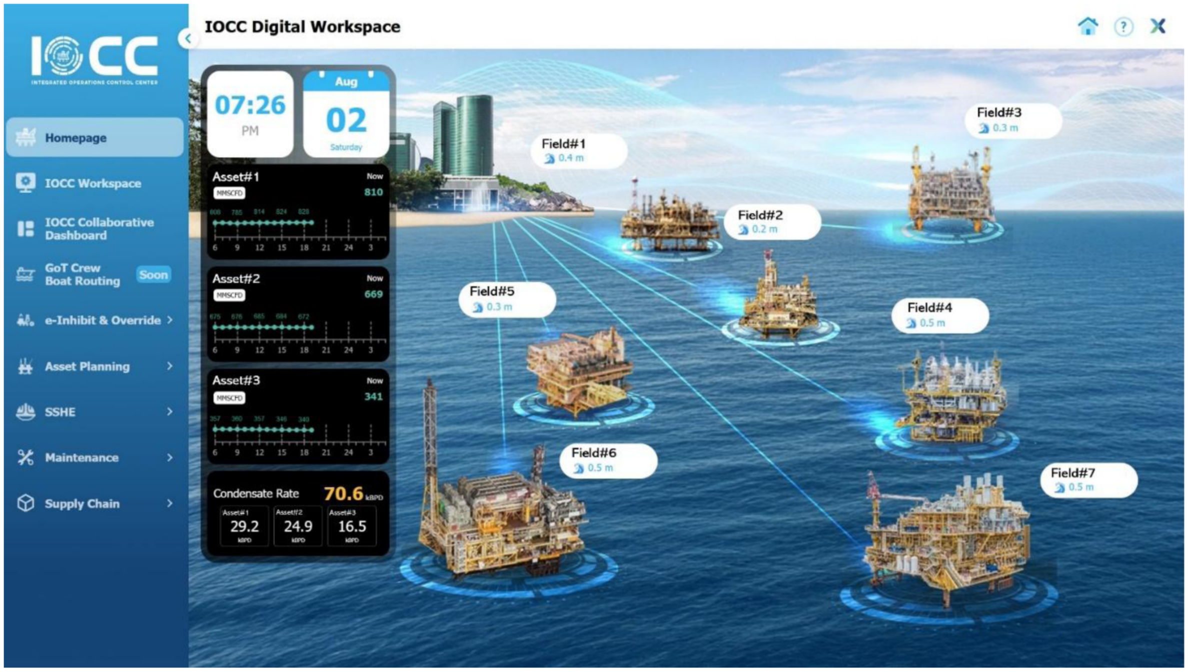

Figure 5—IOCC Homepage, providing a comprehensive overview and access to various critical functions

Main Dashboard for Collaborative Decision-Making and Operational Monitoring

The Main Dashboard is a central component of the IOCC platform, purposefully developed to support collaboration among multi-disciplinary teams involved in day-to-day operations, planning, analysis, and decision-making across various functional domains. It serves as an integrated visual interface where key data—already refined through the three mechanisms of transformation (Visualization and Pre-calculation, Thresholds and Advisories, and User Customization)—are presented in a way that aligns with both strategic and tactical needs. The dashboard is organized into three hierarchical levels, each providing a progressively deeper level of detail, enabling users to access information appropriate to their scope of responsibility and time horizon of decision-making.

Level 0 – Gulf of Thailand (GoT) Production Summary. This top-level dashboard view provides a concise but comprehensive summary of the overall production landscape in the Gulf of Thailand. It is primarily intended to serve the needs of executive management and external stakeholders who require a snapshot of production health without delving into technical complexity. Information presented includes aggregated production volumes for natural gas and condensate, benchmarked against daily sales targets and nominations. In addition, the page visualizes operational control metrics such as key safety, health, and environmental (SHE) statistics, greenhouse gas (GHG) emissions and intensity ratios. These insights are refreshed daily and are aimed at high-level monitoring and trend recognition. The dashboard also includes a toggle function that allows switching to a non-confidential version, specifically tailored for public-facing events or facility visits by guests, ensuring transparency while maintaining data security.

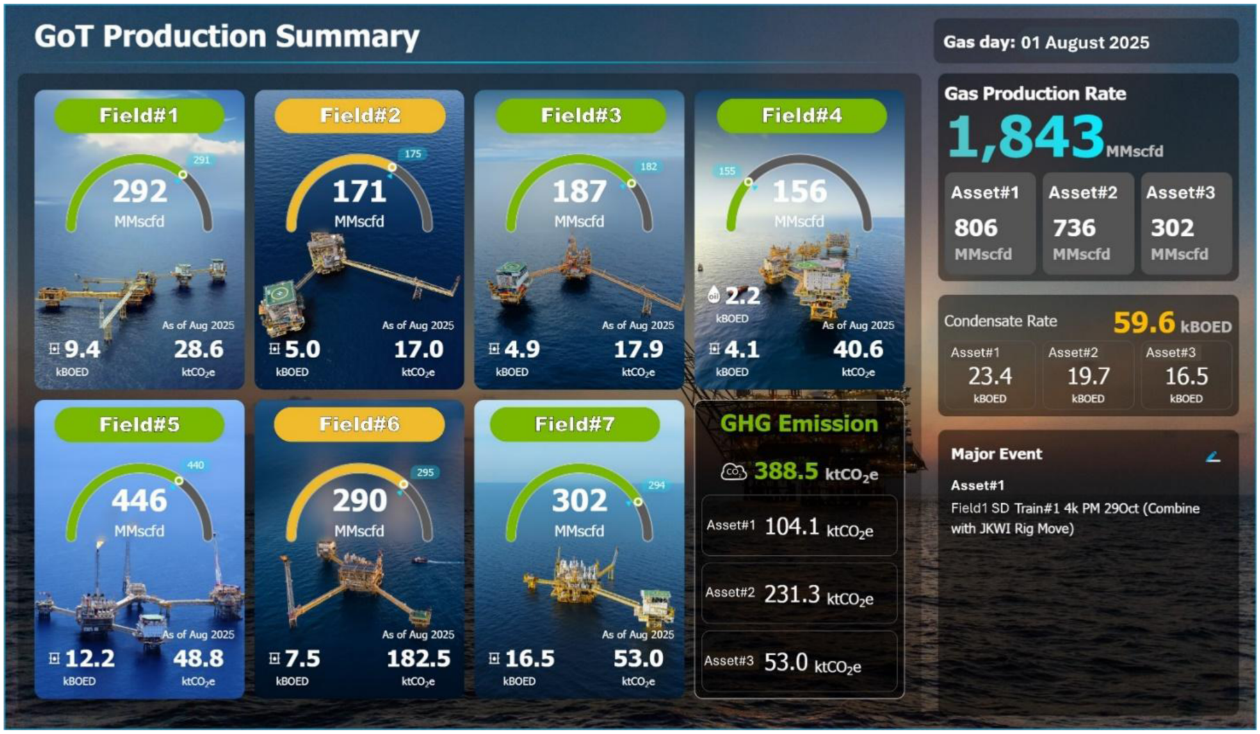

Figure 6—Collaboration Dashboard Level 0: GoT Production Summary Dashboard

Level 1 – Asset Performance Monitoring. This mid-level dashboard focuses on asset-specific and inter-field parameters that are relevant to shared resource utilization, system-wide optimization, and cross-functional collaboration. It is particularly valuable for operational and planning teams who need to monitor ongoing performance and respond dynamically to resource constraints or opportunities. Metrics include, but are not limited to, gas production potential, daily sales volume by asset, number of personnel on board (POB), and accommodation availability. These metrics are essential for managing shared infrastructure and ensuring smooth coordination across different teams. A unique capability of this level is the built-in performance advisory function, which goes beyond displaying historical Key Performance Indicators (KPIs). The system computes forward-looking scenarios that suggest how the remaining operational days of the year must perform to meet predefined annual targets—thereby shifting the perspective from retrospective reporting to proactive planning. This reinforces the core philosophy of the IOCC: it is not only about centralizing operations in a physical space, but also about fostering deeper operational collaboration through shared situational awareness and insight.

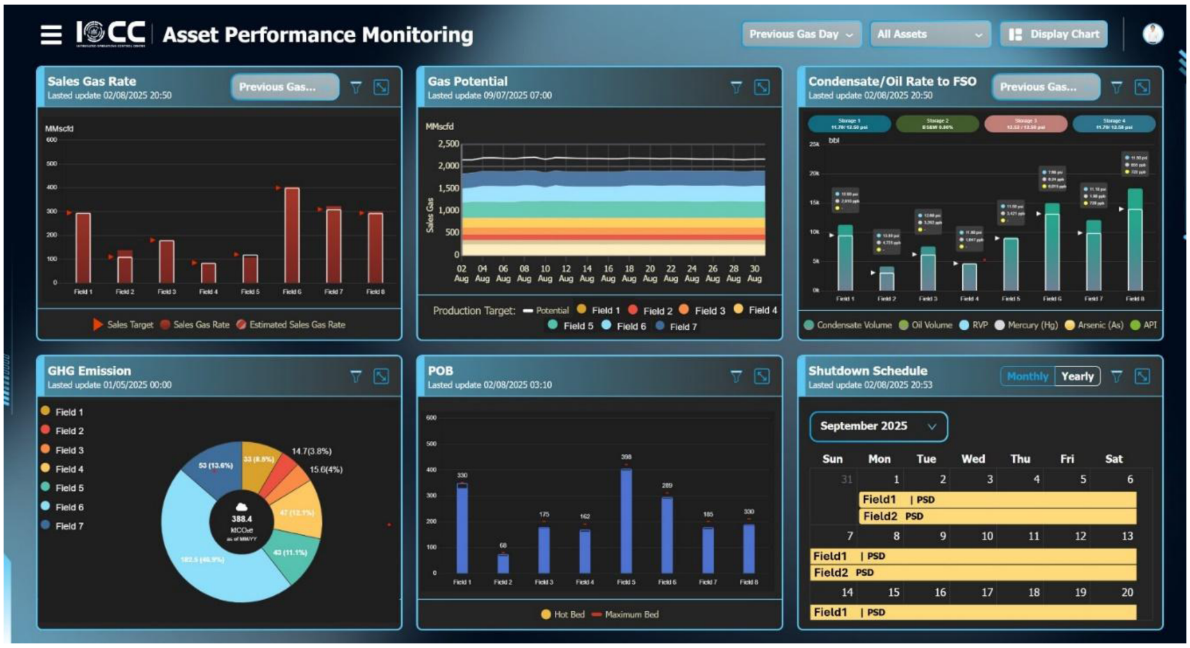

Figure 7—Collaboration Dashboard Level 1: Asset Performance Monitoring Dashboard

Level 2 – Individual Field Information. This granular level offers field-specific dashboards that support daily operational meetings, focused technical discussions, and frontline decision-making. It allows engineers and field operators to drill down into high-resolution data that is directly tied to ongoing operational conditions and potential anomalies. Visualizations at this level include real-time and near-real-time data streams plotted alongside domain-expert-defined thresholds, enabling instant anomaly detection and timely intervention. A strong emphasis is placed on user customizability—users can configure chart layouts, select preferred data views, apply filters for specific parameters, and adjust the time window to suit their analysis needs. Examples of insights shown here include sales gas and condensate volume against specification, corrosion inhibitor injection availability at remote wellhead platforms (which is critical to ensuring pipeline integrity), and GHG-related metrics such as fuel gas usage and flare emissions. This layer also interfaces with underlying AI-powered predictive models; for instance, a model designed to optimize CO_2 removal membrane performance is integrated to suggest operational conditions that minimize hydrocarbon loss while maintaining both product quality and sales volume. In essence, this level serves not just as a dashboard, but as a decision-support tool that bridges raw data, analytics, and field-level actions in real time.

Figure 8—Collaboration Dashboard Level 2: Individual Field Information Dashboard

Workspace for Integrated Operational Activities

The Workspace component of the IOCC platform is designed to support integrated, hands-on collaboration across functional teams by providing a centralized environment for operational planning, troubleshooting, and follow-up activities. It serves as a shared digital workspace where key parameters—initially identified through dashboards—can be further analyzed, discussed, or acted upon in real-time. The goal is to create not only visibility, but also ownership and alignment across roles, empowering cross-disciplinary teams to make data-informed decisions and co-develop solutions in a single, cohesive environment.

Each workspace functions as a collaborative hub for specific domains or processes, combining customizable dashboards to visualize operational data and analytics, maintaining historical context, embedding communication channels directly within operational tasks, and minimizing miscommunication. This makes it easier for frontline personnel, technical engineers, and managers to coordinate activities without siloed tools or fragmented communication. Users can annotate charts, attach relevant documentation, assign responsibilities, and monitor task progression—all while accessing live data streams and related historical trends. The result is not only faster for issue resolution, but also more consistent and transparent operational practices across the entire organization.

At present, the platform supports four foundational work areas, with the flexibility to evolve or expand over time:

Gas Management. Offers centralized tracking and planning of gas production and allocation, ensuring accurate and efficient production management.

Overall Gas Production

Provides real-time visibility of gas production data including sale gas, fuel gas, and flare gas, enabling immediate monitoring and timely problem resolution.

Optimized Well Configuration

Integrates with Potential and Well Testing data from the PDMS system, utilizing advanced Data Analytics driven by in-house algorithms developed by the Subsurface team. This facilitates optimization of over 500 producing wells, immediately increasing Condensate Production.

CO_2 Membrane Optimization

Implements CO_2 membrane optimization using an in-house developed AI-driven models, utilizing real-time data from the PI system to obtain the most up-to-date optimized parameters, significantly reducing hydrocarbon loss

Gas Ramp up/down Simulation

Simulates gas ramping scenarios during plant turnarounds by linking real-time gas production data with daily nomination inputs from the Sales team, enhancing the effectiveness of production planning.

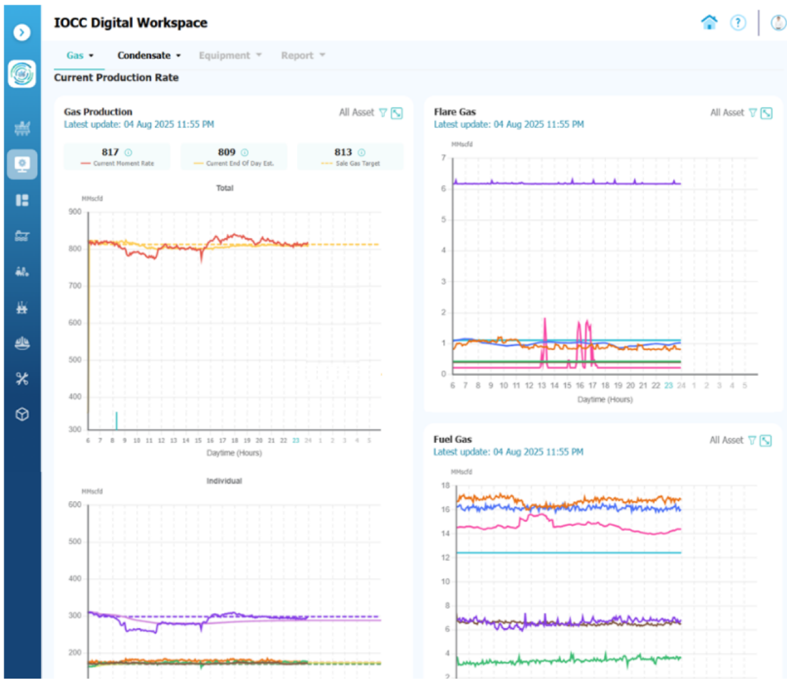

Figure 9—Gas Management Workspace, A centralized tracking and planning of gas production and allocation.

Condensate Management: Monitoring of condensate handling, quality, and storage management.

The condensate management workspace creates an **end-to-end** collaborative workspace that streamlines processes starting from receiving production targets set by the Sales Commercial team through to the preparation of Condensate Production data by Petroleum Engineering. In addition, this workspace provides the ability to simulate condensate production allocation, integrating quality data from laboratory results and quantity data from PI data sources, while managing storage constraints imposed by shared floating storage tanks across multiple production fields.

This comprehensive design promotes cross-team collaboration and provides a trusted single data source, reducing work steps, minimizing errors, and improving overall efficiency.

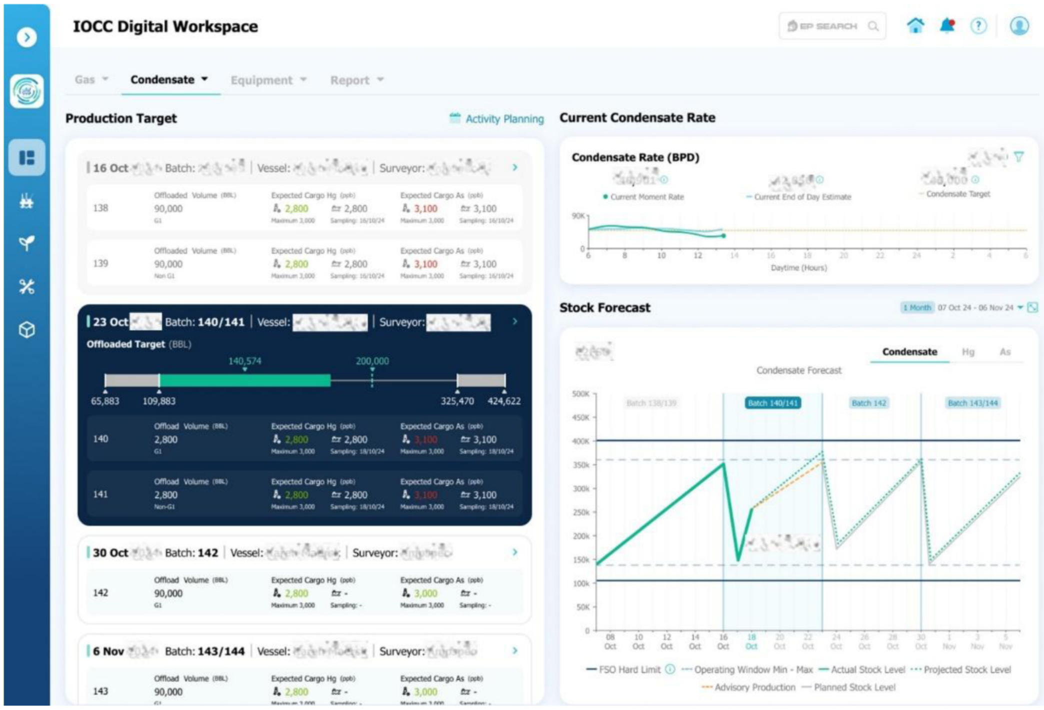

Figure 10—Condensate Management Workspace, an end-to-end working collaboration platform for condensate management.

Booster Compressor & Equipment Monitoring: Real-time condition monitoring of rotating equipment. This workspace connects real-time status updates of Booster Compressors and Cranes located on Wellhead Platforms, utilizing the latest maintenance data to support efficient operational decision-making within the IOCC team. It includes data integration from third-party equipment surveillance systems, assisting in proactive planning and timely replacement of consumable parts such as filters or absorbents in treatment systems to minimize downtime and optimize equipment reliability.

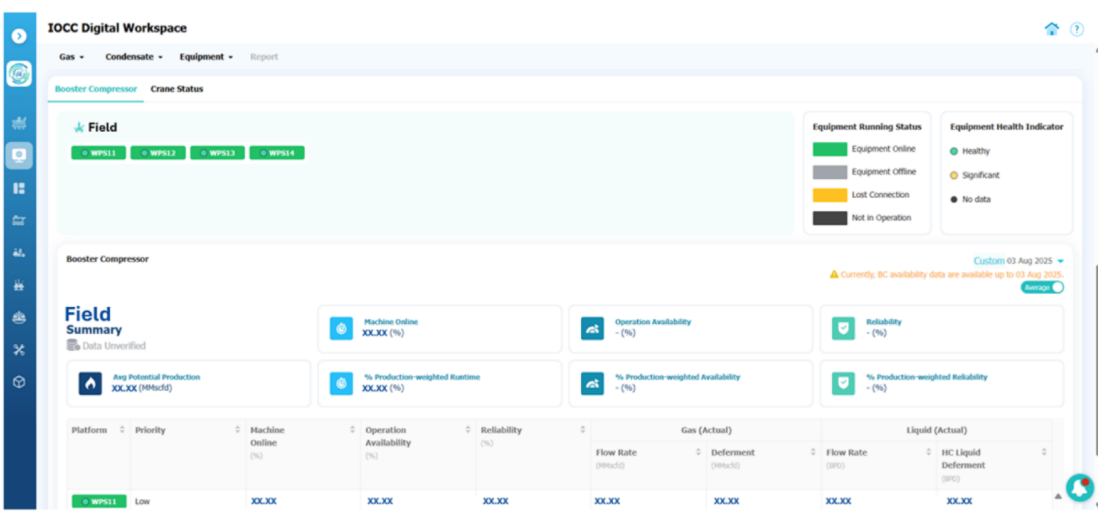

Figure 11—Real-time Monitoring of Booster Compressors and Equipment Workspace

Operational Reports Automation: Digitized daily and shift-based reporting for traceability and efficiency. This workspace streamlines the creation of critical operational reports such as Shutdown Reports and Production Reports, which traditionally involve extensive data compilation. It facilitates easy access to necessary data sources, automates report generation processes, and incorporates approval workflows with notifications to relevant stakeholders, significantly enhancing report accuracy and accelerating the reporting cycle time from initial drafting to final submission.

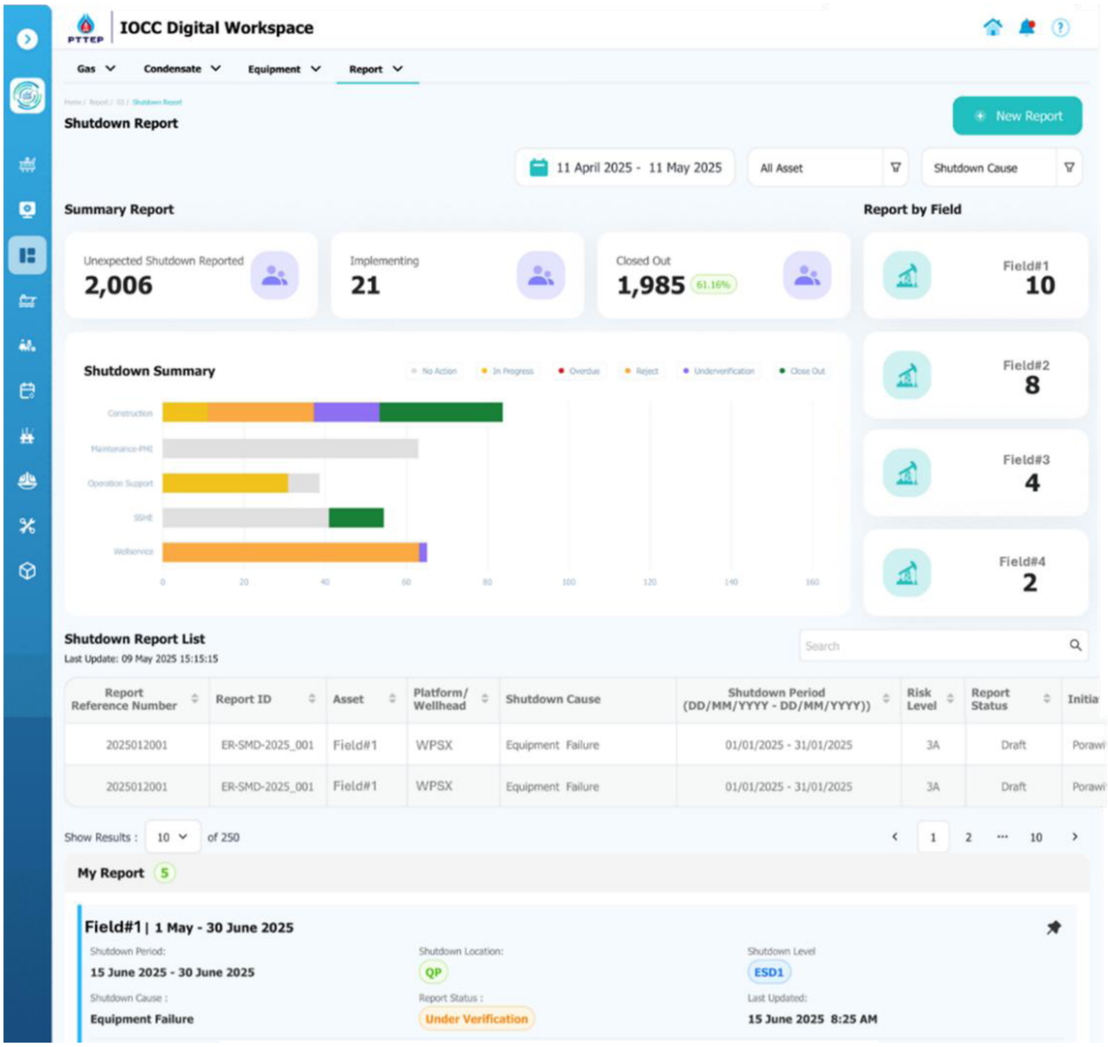

Figure 12—Automated Operations Reporting System Workspace

By aligning these key features with the operational processes of the operations and related support function teams, Workspace enables enhanced efficiency, effective real-time collaboration, and data-driven decision-making within IOCC Digitalization.

Operational Applications

The Operational Applications segment consolidates essential shortcut access to various critical operational tools and applications, integrated seamlessly through a robust Data Foundation. This foundation ensures efficient data management, eliminates data redundancies, and significantly reduces data silos, enabling optimized and streamlined operations within IOCC. Examples of key operational applications include:

e-Inhibit & Override. A comprehensive workflow management system critical for IOCC operations, especially with control room panel operators stationed at the Bangkok headquarters. This centralized digital platform effectively manages the inhibit and override processes for field devices, replacing traditional manual, paper-based workflows. The system integrates seamlessly with the e-PTW (Permit to Work) and MOC (Management of Change) databases, enabling instant attachment and verification of relevant documentation, enhancing operational compliance and safety.

Logistic Fleets Management. A pivotal digital solution supporting Marine Control teams in Offshore Vessel Management, optimizing vessel operations within the GoT. This application streamlines transportation logistics for operational crews, contractors, and materials with AI capability to recommend efficient route based on input criteria, operational constraints, current demand, and real-time data. It also features real-time Vessel Tracking capabilities for immediate status updates and prompt decision-making. Includes Crew Boat Route Planning with advanced Route Optimization features, enhancing daily operational efficiency by systematically planning crew transportation routes to various wellhead platforms.

Integrated Operations Activities Planning. A single-database, web-based planning system facilitating integrated offshore activity management. It provides consolidated visibility for various planning horizons, including 4-week Lookahead, 3-month Lookahead, and 1-Year Lookahead schedules. Enables all functional planners and Integrated Operations Planners to utilize a unified Integrated Activities Planning (IAP) platform, significantly reducing planning errors and improving accuracy and responsiveness. This application Direct integration with Crew Boat Applications further enhances operational efficiency, coordination, and daily planning reliability.

In addition to these key applications, IOCC Digitalization remains agile and adaptable, actively exploring and integrating emerging digital technologies such as Agentic AI, to continually maximize operational efficiency and effectiveness.

Figure 13—Logistic Fleet Management Application

Results

The IOCC digital solution is being deployed through a phased rollout, with Phase 1 targeted for Aug'2025 alongside the relocation of offshore teams to the centralized command center in Q4.

Key outcomes include improved real-time monitoring of production and field operations, more precise control over wellhead platforms, and optimized CO_2 membrane performance. The introduction of GHG tracking tools has also strengthened environmental oversight through more timely and accurate data.

The centralized IOCC setup has enhanced cross-asset collaboration across three offshore zones, comprising seven production fields and three floating storage units. Standardized dashboards and shared digital workspaces have allowed multidisciplinary teams to respond more quickly and operate with better situational awareness.

Notably, the IOCC digitalization is projected to deliver an estimated **USD 3.2 million in annual value realization** through improved operational efficiency, reduced equipment downtime, better resource allocation, and data-driven decision-making. This value stems not only from cost savings but also from productivity gains, risk mitigation, and faster time-to-action enabled by a unified digital environment.

Figure 14—Rollout timeline for IOCC Digitalization initiative.

Discussion

The IOCC's early impact highlights not just technological advancement but also a shift in operational thinking. Centralizing control functions into a single platform reduces fragmentation between field units, technical teams, and management—allowing for a unified, real-time operational view that supports faster and more cohesive decisions.

This transformation also breaks down functional silos. Live data access and contextual insights support continuous collaboration across disciplines, leading to higher-quality responses and better alignment in both routine operations and urgent scenarios.

More importantly, the platform encourages a proactive mindset. Forecasting tools and advisory logic embedded in daily workflows allow users to identify issues before they escalate—building resilience into PTTEP's operations.

However, realizing long-term value will require more than digital tools. Sustaining adoption demands clear workflow ownership, ongoing training, and a strong feedback loop for iterative improvement. As the system grows more interconnected, data integrity, cybersecurity, and redundancy planning will be critical—not just in the platform's architecture, but also in governance frameworks and operational practices.

Conclusions and Recommendations

The IOCC Digitalization project represents a significant milestone in PTTEP's long-term strategy to achieve intelligent, integrated offshore operations. The project has already demonstrated measurable operational improvements, including minimized equipment downtime, enhanced inter-departmental collaboration, and more effective environmental monitoring. These early successes validate the direction of the initiative and underscore the potential for even broader organizational impact.

As the implementation phase progresses, several key recommendations can be made to maximize long-term value:

Predictive analytics: Expand the system's analytical capabilities to predict potential equipment failures and maintenance needs, reducing unplanned shutdowns and improving asset reliability.
Digital twin integration: Implement virtual replicas of offshore assets to simulate scenarios, test response strategies, and support proactive decision-making in both operations and safety planning.
Executive-level dashboards: Develop real-time KPI dashboards tailored for leadership to support strategic oversight, faster decision-making, and transparent performance tracking.
User-centric evolution: Foster a culture of continuous improvement by incorporating end-user feedback into the platform's ongoing development. This ensures relevance, ease of use, and broader adoption.
Strong data governance: Establish robust data governance frameworks to ensure data quality, security, and compliance, which are critical for scaling the platform enterprise-wide.
Advanced AI and Agentic AI Implementation**:** Integrate advanced Artificial Intelligence (AI) and Agentic AI capabilities to move beyond predictive to be proactive analytics towards autonomous decision-making, continuous optimization and self-Improvement and proactive operational adjustments.

Together, these enhancements can help transform the IOCC from a functional innovation into a long-term strategic for PTTEP.

Novelty and Contributions

The IOCC project goes beyond standard digital upgrades seen in the energy sector. Unlike conventional offshore operations that typically rely on fragmented control systems and siloed data, this initiative introduces a holistic digital transformation framework that integrates control, analytics, and collaboration tools into a unified platform.

This integrated approach enables cross-functional visibility and decision-making, which is particularly novel in high-stakes, real-time offshore environments. Among the key innovations are:

- The use of **intelligent dashboards** that consolidate real-time operational data from multiple sources.
- The deployment of **digital workspaces** that enable synchronous end-to-end collaboration across teams located onshore and offshore.
- The facilitation of **data-driven operations** that empower operators and executives alike to act quickly and confidently, with a shared understanding of asset performance.

This paper provides actionable insights into how oil and gas companies, especially national or regional operators, can accelerate their digital transformation. It also offers a replicable model for integrating smart technologies to enhance safety, efficiency, and sustainability in offshore operations.

References

This report/document is based entirely on internal knowledge, experiences, observations, and development activities. *No external sources were referenced*.